

# TRT HORMONE OPTIMIZATION GUIDE

## Ch 1: Understanding TRT

Testosterone Replacement Therapy restores testosterone to physiological levels in men with clinically diagnosed hypogonadism. At medical TRT doses, the goal is wellness restoration — not supraphysiological performance. Understanding this distinction is critical.

| TRT Category      | Description                                                            | Target Total T Level                     |
|-------------------|------------------------------------------------------------------------|------------------------------------------|
| Medical TRT       | Prescribed by doctor; restores to normal physiological range           | 400–700 ng/dL                            |
| Optimised TRT     | Slightly above average normal; enhanced wellbeing and body composition | 700–1000 ng/dL                           |
| TRT + ancillaries | HCG added for fertility; AI for E2 control; full monitoring protocol   | 700–900 ng/dL with normal LH/FSH via HCG |

## Ch 2: Protocol Options

| Delivery Method                  | Dose             | Frequency                 | Advantages                                | Disadvantages                                  |
|----------------------------------|------------------|---------------------------|-------------------------------------------|------------------------------------------------|
| Testosterone Cypionate injection | 100–200 mg/mL    | Weekly or E3.5D           | Cost-effective; consistent levels; stable | Requires injection technique; equipment needed |
| Testosterone Enanthate injection | 100–200 mg/mL    | Weekly or E3.5D           | Well-established; widely available; cheap | Same as TC; slight peak-trough variation       |
| Testosterone gel                 | 50–100 mg/day    | Daily topical application | No injections; easy; stable levels        | Transfer risk to family; variable absorption   |
| Testosterone pellets             | 150–450 mg total | Every 3–6 months          | Very infrequent; consistent levels        | Surgical implant; inflexible dose adjustment   |
| Testosterone undecanoate         | 1000 mg/3 mL     | Every 10 weeks            | Longest duration; infrequent              | Expensive; requires clinic administration      |

## Ch 3: Blood Work Reference

| Marker             | Optimal Range on TRT        | Action if Out of Range                                            |
|--------------------|-----------------------------|-------------------------------------------------------------------|
| Total Testosterone | 600–900 ng/dL               | Adjust dose; check timing of blood draw (always at trough)        |
| Free Testosterone  | 15–25 pg/mL                 | May need SHBG management with Boron 10mg or Proviron              |
| Estradiol (E2)     | 20–35 pg/mL                 | Add AI if > 40; reduce AI if < 15; retest in 4 weeks              |
| SHBG               | 20–40 nmol/L                | Low SHBG: more frequent smaller injections; high: standard dosing |
| Haematocrit        | < 52%                       | Donate blood if > 52%; reduces clot risk; check every 3 months    |
| LH / FSH           | Near zero (expected on TRT) | Add HCG if fertility is a concern                                 |
| PSA                | < 2.0 ng/mL                 | Annual check; consult urologist if rapidly rising                 |
| HDL Cholesterol    | > 40 mg/dL                  | Omega-3, cardio, statin if needed; TRT can lower HDL              |
| ALT / AST (liver)  | < 2x upper limit of normal  | Injectable TRT has minimal liver impact; switch if using oral     |

# Ch 4: HCG on TRT

HCG (Human Chorionic Gonadotropin) mimics LH and maintains testicular function, volume, and intratesticular testosterone while on TRT. It is essential for men who wish to preserve fertility on long-term TRT.

| HCG Use Case                     | Dose        | Frequency                  | Benefit                                          |
|----------------------------------|-------------|----------------------------|--------------------------------------------------|
| Fertility preservation on TRT    | 500–1000 IU | 3x/week                    | Maintains sperm production; prevents azoospermia |
| Testicular atrophy prevention    | 250–500 IU  | 2–3x/week                  | Prevents testicular volume loss during TRT       |
| Restart pre-PCT (if cycling)     | 500 IU      | EOD for 4 weeks before PCT | Kickstarts HPTA before SERM therapy              |
| Low libido despite good T levels | 500 IU      | 2x/week                    | Raises intratesticular testosterone              |

## Ch 5: Lifestyle Optimisation on TRT

### Maximise TRT Results Through Lifestyle

- Resistance training 3–5x/week: amplifies the anabolic signal of exogenous testosterone
- Sleep 7–9 hours: GH release and synergy with testosterone occur during deep sleep
- Body fat: keep BF under 20%; adipose tissue converts testosterone to estrogen via aromatase
- Alcohol: limit to under 2 units/week; alcohol suppresses T and raises estrogen
- Stress: chronic cortisol impairs testosterone receptor sensitivity and reduces free T
- Micronutrients: Zinc 30mg + Magnesium 400mg + Vitamin D 5000 IU support T pathway enzymes
- Avoid xenoestrogens: BPA plastics, pesticides, and certain personal care product ingredients

| Common TRT Concern           | Explanation                              | Solution                                                    |
|------------------------------|------------------------------------------|-------------------------------------------------------------|
| Polycythaemia (high RBC/HCT) | TRT stimulates red blood cell production | Donate blood every 2–3 months; adjust dose if persistent    |
| Testicular atrophy           | LH suppression shrinks testes over time  | Add HCG 250–500 IU 2–3x/week; usually reverses within weeks |
| Fertility impairment         | Exogenous T suppresses sperm production  | HCG + FSH (or Clomid) can maintain sperm production on TRT  |
| Sleep apnoea worsening       | TRT can worsen upper airway muscle tone  | CPAP if diagnosed; reduce TRT dose if correlated            |
| Acne / oily skin             | Androgen effect on sebaceous glands      | Benzoyl peroxide wash; zinc; lower dose if severe           |